FIPS 140-3 Accredited Lab Selection — pqc_crypto_module v0.1.0

> **Disclaimer**: Prepared for FIPS 140-3 evaluation, not currently validated.

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1. Candidate Labs

1.1 atsec Information Security

- **Location**: Austin, TX, USA (with offices in Germany)
- **Website**: https://www.atsec.com
- **FIPS 140-3 Experience**: Long-standing NVLAP-accredited lab; extensive FIPS 140-2 and 140-3 validation history across software, firmware, and hardware modules.
- **PQC Experience**: Active in standards evolution; positioned for FIPS 203/204 algorithm testing as NIST finalizes ACVP support.
- **Estimated Responsiveness**: High. Known for structured onboarding and clear communication.
- **Notes**: Strong reputation in open-source and Linux-based module validation. Good fit for Rust/software-only modules.

1.2 UL Solutions (formerly UL Verification Services / InfoGard successor programs)

- **Location**: Northbrook, IL, USA (global presence)
- **Website**: https://www.ul.com
- **FIPS 140-3 Experience**: One of the largest NVLAP-accredited labs with hundreds of completed validations.
- **PQC Experience**: Broad cryptographic algorithm testing capabilities; expected early adoption of PQC CAVP testing.
- **Estimated Responsiveness**: Medium-High. Large lab with multiple concurrent engagements; scheduling may vary.
- **Notes**: Full-service lab covering FIPS 140-3, Common Criteria, and PCI. Suitable for vendors seeking multiple certifications.

1.3 Acumen Security

- **Location**: McLean, VA, USA
- **Website**: https://www.acumensecurity.com

- **FIPS 140-3 Experience**: Specialized FIPS 140-3 testing lab with deep expertise in software cryptographic modules.
- **PQC Experience**: Actively tracking NIST PQC standards; positioned for early ML-DSA/ML-KEM validation support.
- **Estimated Responsiveness**: High. Smaller lab with focused attention per engagement.
- **Notes**: Known for working closely with vendors on documentation and design alignment. Good fit for first-time CMVP submitters.

1.4 Leidos (Cyber Innovation Center)

- **Location**: Reston, VA, USA
- **Website**: https://www.leidos.com
- **FIPS 140-3 Experience**: NVLAP-accredited lab with government and defense sector validation experience.
- **PQC Experience**: Government contracts position them close to NIST PQC transition timelines.
- **Estimated Responsiveness**: Medium. Larger organization; engagement timelines depend on current workload.
- **Notes**: Strong government sector relationships. Appropriate if targeting US federal procurement channels.

1.5 InfoGard Laboratories

- **Location**: San Luis Obispo, CA, USA
- **Website**: https://www.infogard.com
- **FIPS 140-3 Experience**: Established NVLAP-accredited lab with decades of FIPS validation experience.
- **PQC Experience**: General cryptographic testing; PQC-specific CAVP testing readiness to be confirmed.
- **Estimated Responsiveness**: Medium-High. Mid-sized lab with dedicated validation teams.
- **Notes**: Well-known in the CMVP community. Solid choice for straightforward software module validations.

2. Selection Criteria

Criterion	Weight	Description
FIPS 140-3 Track Record	High	Number of completed FIPS 140-3 validations, especially software modules
PQC Readiness	High	Ability to test ML-DSA-65 (FIPS 204), ML-KEM-768 (FIPS 203), SHA3-256 (FIPS 202)
Rust Module Experience	Medium	Prior experience validating Rust-based cryptographic modules
Communication Quality	Medium	

Criterion	Weight	Description
		Responsiveness, clarity of feedback, structured onboarding process
Timeline Predictability	Medium	Ability to provide and adhere to estimated schedules
Cost Transparency	Medium	Clear pricing structure; no hidden fees for iterations
Geographic Accessibility	Low	Timezone alignment for meetings; in-person visits if needed

3. Recommended Approach

1. **Contact 2-3 labs** with the executive summary and module overview from `CONTACT_PACKAGE/`.
2. **Request**: estimated timeline, cost range, PQC algorithm testing readiness, and any Rust-specific considerations.
3. **Evaluate** responses against the criteria above.
4. **Select** based on PQC readiness and communication quality as primary differentiators, given the novelty of ML-DSA/ML-KEM validation.

4. Initial Contact Priority

Priority	Lab	Rationale
1	Acumen Security	Specialized focus, vendor-friendly, high responsiveness
2	atsec	Strong open-source track record, structured process
3	UL Solutions	Scale and breadth; fallback if top choices unavailable